

Original software publication

# OSP-SAP: A MATLAB graphical user interface for optimal sensor placement using SAP2000

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## ABSTRACT

An optimal placement of sensors is key for structural modal identification since it critically determines the quality of the estimated modal properties. To facilitate this task, this paper presents OSP-SAP, a MATLAB-based graphical user interface designed for Optimal Sensor Placement (OSP) using SAP2000 finite element models. The software integrates four different optimization algorithms to determine optimal sensor configurations while providing intuitive visualization tools. Additionally, OSP-SAP streamlines practical implementation by allowing users to export the generated optimal configurations to AutoCAD and generate Word reports, making OSP more accessible to researchers and practitioners.

## Code metadata

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## 1. Motivation and significance

Structural Health Monitoring (SHM) is gaining increasing interest in practice as an effective means to ensure the safety and longevity of infrastructure. Among the existing techniques, vibration-based SHM utilizing Operational Modal Analysis (OMA) has become particularly popular due to its non-destructive nature and minimal intrusiveness. This approach estimates the modal properties of structures (i.e., natural frequencies, mode shapes, and damping ratios) as damage-sensitive features from their ambient vibration response, without exerting artificial excitations [1]. However, a key challenge in achieving accurate modal identification results is the proper placement of sensors, which is particularly critical when only a limited number are available. To this end, a number of OSP algorithms have been proposed in the literature.

These techniques seek to determine the  $l$  optimal sensor locations among  $n$  accessible positions that optimize a quality metric defined by the specific OSP algorithm.

Since OSP was first introduced [2], numerous algorithms have been proposed in the literature [3]. Recent advances have expanded traditional OSP methods to address model uncertainties [4,5], include advanced evaluation criteria [6,7], incorporate multi-type sensors [8], and include the effect of loading conditions on optimal placement [9], among others. Despite these developments and the existence of extensive validation in the literature [10–12], there remains a notable lack of software solutions for practical implementation in routine engineering workflows.

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To the best of the authors' knowledge, only two software tools for OSP supported by Graphical User Interfaces (GUIs) have been reported in the literature. Yi et al. [13] developed an OSP strategy encapsulated in a MATLAB toolbox to determine the optimal number and location of sensors, although their toolbox is not openly available. Nicoletti and co-authors [14] also developed a MATLAB-based GUI incorporating five algorithms to solve the OSP problem. Despite its potential, this tool lacks support for multi-setup configurations (commonly required for modal testing of large structures), does not allow exporting sensor layouts to CAD software, and requires manual input of Finite Element Analysis (FEA) data—a common critical step in most available OSP tools.

To address these gaps, this work introduces OSP-SAP, a MATLAB-based GUI [15] integrated with SAP2000 [16] that automates the OSP workflow through a user-friendly interface. Unlike previously reported software tools, OSP-SAP enables users to work directly with Finite Element Models (FEMs) within a unified workflow, without the need to manually extract intermediate analysis results such as mode shapes or stiffness and mass matrices. SAP2000 has been selected as the FEM solver due to its widespread use in both engineering and academic practice, including several studies addressing OSP problems [4, 17–19]. OSP-SAP implements four well-established OSP algorithms (partly implemented by some of the authors in [10–12,20]), namely: (i) Effective Independence (EFI) [21], (ii) Mass-Weighted Effective Independence (EFIwm) [22], (iii) Strain Energy Matrix Rank Optimization (SEMRO) [23], and (iv) Kinetic Energy Matrix Rank Optimization (KEMRO) [24]. A concise description of each of them is provided below:

- **EFI** places sensors to maximize the linear independence of the mode shapes, thereby enhancing their observability. Among the four algorithms, EFI is the least computationally burdensome.
- **EFIwm** extends EFI by incorporating mass weighting, making it particularly suitable for FEMs with significant variations in mass distribution.
- **SEMRO** places sensors at locations that maximize the strain energy of the identified mode shapes. It is particularly suitable for damage detection applications, as structural damage typically alters the strain energy distribution of modes mobilizing damaged elements [25].
- **KEMRO** prioritizes sensor placement at locations with the highest kinetic energy, which is particularly recommended for maximizing the signal-to-noise ratio [26].

It is important to note that these algorithms are iterative and greedy, and thus provide suboptimal solutions without guaranteeing a globally optimal sensor placement [21,27]. Furthermore, uncertainties in the utilized FEM can affect the accuracy of the resulting sensor configurations [4,10]. Users are therefore advised to interpret the results with caution, especially in practical applications where such uncertainties are unavoidable.

## 2. Software description

OSP-SAP has been developed in the MATLAB environment as a compact GUIDE Graphical User Interface (GUI). The software code is available in [GitHub](#) and [Zenodo](#) repositories, and can be executed by simply running the `Run_Interface.m` file. Alternatively, it can be installed as a MATLAB App by double-clicking the `OSP-SAP.mlappinstall` file or using the `Install App` button in MATLAB.

Once installed, users must input a FEM model from SAP2000, and the rest of the OSP workflow is controlled within OSP-SAP, as depicted in Fig. 1.

The software addresses two types of OSP problems:

- **Global OSP (G-OSP)**, aimed at determining the optimal placement of a user-defined number of sensors from a single set of candidate DOFs.
- **Multi-setup OSP (M-OSP)**, which optimizes the placement of roving sensors across multiple groups (setups) of candidate DOFs while keeping some sensors as fixed references. This approach is used in multi-setup modal identification scenarios, where a limited number of sensors prevents full-structure monitoring in a single test. In this case, the structure is divided into setups, defined as subgroups of candidate nodes, with one test conducted per setup. Reference sensors remain fixed throughout all tests, while roving sensors are repositioned according to each setup. Internally, M-OSP is solved by applying the OSP algorithm separately to each group formed by the reference sensors and the candidate nodes within each setup.

The results, displayed via interactive visualizations, can be refined and exported as AutoCAD layouts or Word reports (see Section 2.2.4).

### 2.1. Software architecture

The OSP-SAP GUI is structured into three main sections, as depicted in Fig. 2:

- **SAP2000 Panel:** This section contains three input fields for specifying the paths to the SAP2000 model database, as well as the paths to the `.exe` and `.dll` files for SAP2000.
- **Visualization Panel:** This section displays the model geometry, mode shapes, and the OSP convergence diagram.
- **OSP Panel:** This section allows selecting and running the OSP algorithm. It comprises three tabs:
  - *Candidate Modes:* Displays a table with the natural vibration frequencies from the FEM and allows the user to select those modes intended for identification (i.e., target modes).
  - *OSP DOFs:* Enabled after running the OSP algorithm, this tab contains a table detailing the optimal sensor positions for the different setups considered in the analysis, allowing the user to define the desired number of sensors. If a G-OSP problem is chosen, this tab optionally allows the user to specify a desired number of reference sensors. While reference sensors are not required to solve the G-OSP problem itself, this information can be used if the user intends to subsequently perform an M-OSP analysis.
  - *Info:* Provides a table containing information about the FEM, including the coordinates of the nodes, number of setups, and reference sensors.

The software workflow follows a sequential process, as illustrated in Fig. 1. The input is provided through a SAP2000 FEM in the **SAP2000 Panel**, as described in Section 2.2.1. Subsequently, the **OSP Panel** is used to run the modal analysis of the previously introduced FEM, select the DOFs and target modes, and execute the chosen OSP algorithm (as described in Section 2.2.3). Finally, the results can be exported in Word and AutoCAD formats (more details are provided in Section 2.2.4). Throughout this process, all intermediate and final results are displayed in the **Visualization Panel**.

### 2.2. Software functionalities

#### 2.2.1. Preparation of FEM in SAP2000

The initial step involves preparing the SAP2000 FEM, which constitutes the main software input, as depicted in Fig. 1. To ensure compatibility, at least one of the following node groups must be defined:

- ‘‘Candidate\_DOFs’’ for G-OSP problems: This group contains the nodes where sensors can be physically installed.

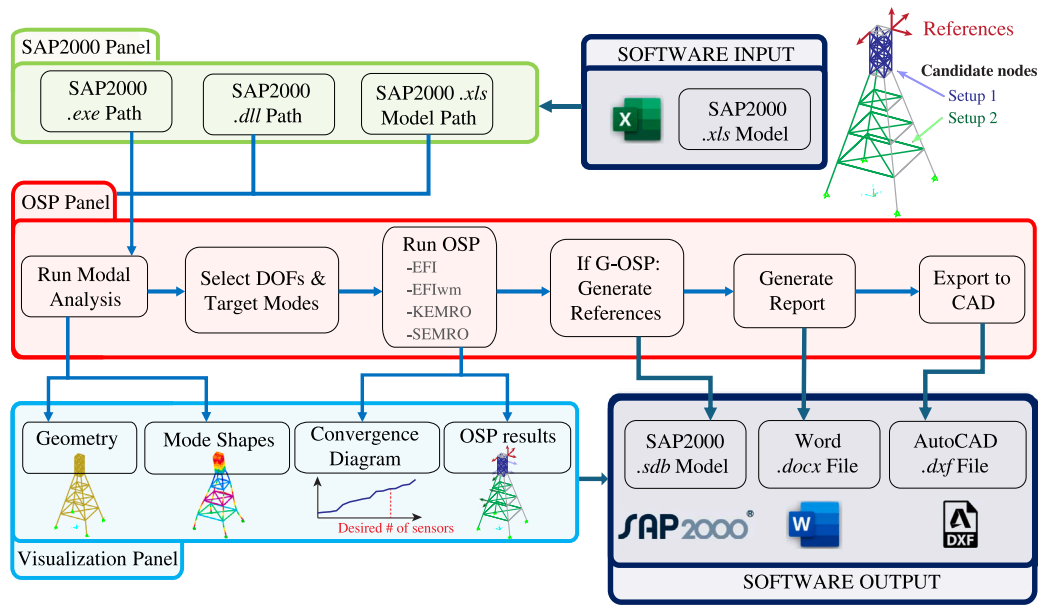


Fig. 1. Workflow of OSP-SAP.

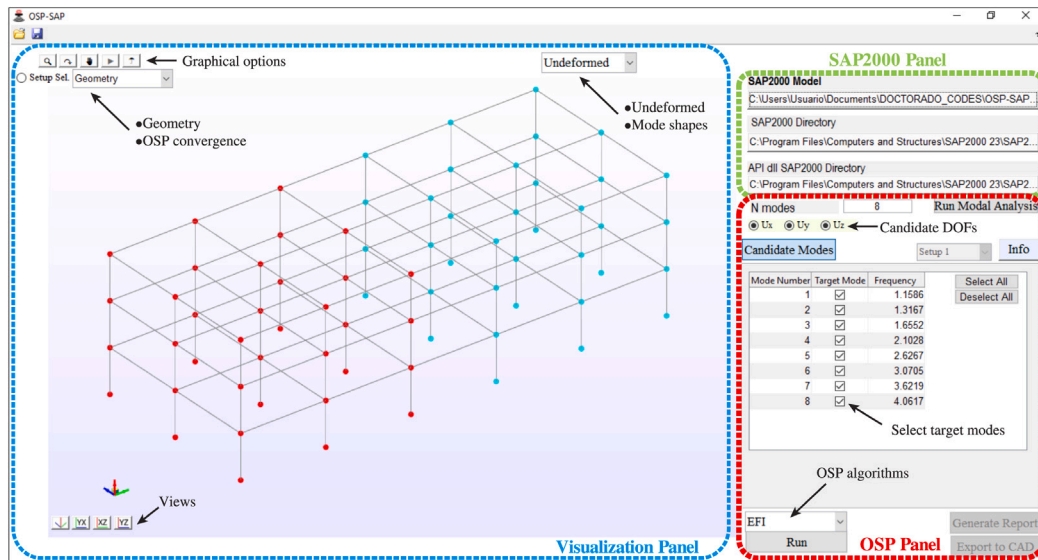


Fig. 2. Layout of the main GUI in OSP-SAP.

- ‘‘Setup\_i’’ (e.g., ‘‘Setup\_1’’, ‘‘Setup\_2’’, ...) for M-OSP problems: Specifies the nodes where sensors can be placed in a particular setup (see Fig. 3(a)).

It is important to highlight that the quality of the OSP results strongly depends on the number and spatial distribution of the candidate nodes. A larger, well-distributed set typically leads to more informative sensor configurations. However, these nodes must also be physically accessible to ensure the practical feasibility of sensor installation.

In addition to the ‘‘Setup\_i’’ groups, an M-OSP problem requires defining a load pattern named ‘‘References’’. This pattern consists of forces applied at the nodes where the reference sensors are located, with their direction matching the measurement direction of the sensors.

Once the SAP2000 model is properly configured, it must be exported as an Excel database (see Fig. 3(b)). The ability to perform G-OSP or M-OSP problems depends on the FEM input: a G-OSP can always be

executed, whereas an M-OSP requires the presence of ‘‘Setup\_i’’ groups and a ‘‘References’’ load pattern. The different scenarios are illustrated in Fig. 4.

### 2.2.2. Load SAP2000 model and run the modal analysis

To import the SAP2000 in OSP-SAP, the user must enter the path to the SAP2000 .xls file in the **SAP2000 Panel** (see Fig. 2) and, if necessary, modify the paths for the SAP2000 .exe and .dll (typically located in C:/Program Files/Computers and Structures/SAP2000 v, where v corresponds to the SAP2000 version). Once defined, the **OSP Panel** allows the user to define the number of mode shapes to be extracted, and the *Run Modal Analysis* button automatically executes the modal analysis of the FEM via the SAP Open Application Programming Interface (OAPI). Upon completion, the software retrieves information about the FEM (i.e., nodes, beam elements, stiffness and mass matrices) as well as the results of the modal analysis (i.e., resonant frequencies and mode shapes), which are necessary for the subsequent OSP.

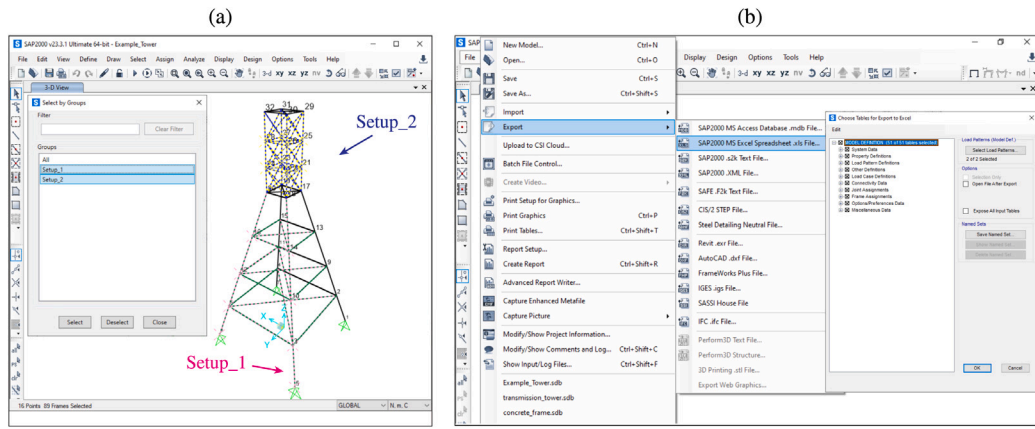


Fig. 3. Definition of node groups for multi-setup analyses (a), and screenshot of the process to export SAP2000 models into .xls databases (b).

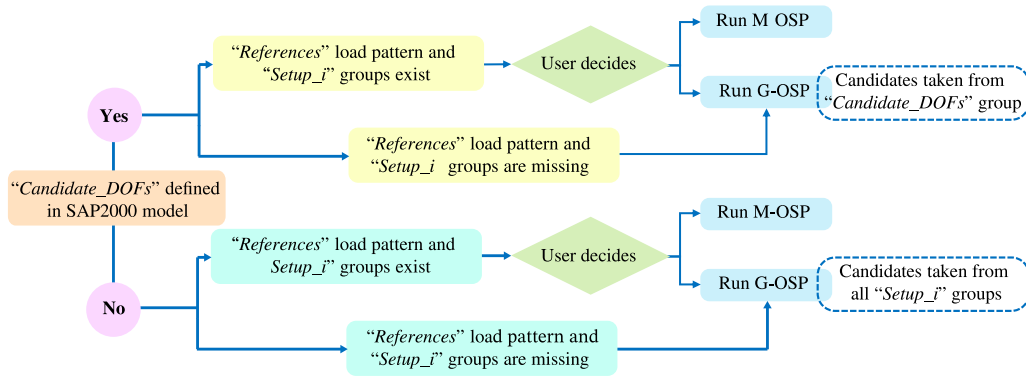


Fig. 4. Functionalities of OSP-SAP depending on the node groups defined in the input FEM.

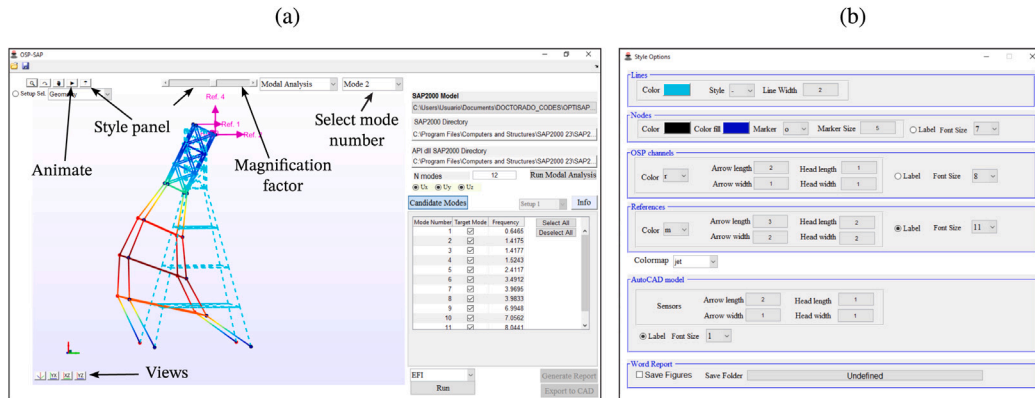


Fig. 5. Screenshot of OSP-SAP displaying a mode shape (a) and style panel (b).

The geometry of the model is displayed in the **Visualization Panel**, where reference sensors and setup groups are also represented. To visualize the model's mode shapes, users can navigate to the upper-right drop-down menu within the panel and select *Modal Analysis* (see Fig. 5(a)). The upper-left section of the panel features buttons for interacting with the visualization. From left to right, these include a zoom loupe, a rotation arrow, a pan hand tool, a play button for animating mode shapes, and an umbrella icon to open the style panel with options to customize the graphical display (see Fig. 5(b)). Finally, the bottom-left section of the panel contains a set of buttons that allow users to change the viewing perspective.

### 2.2.3. Executing the OSP algorithm

Once the modal analysis has been completed, the next step is to define the parameters for solving the OSP problem. Firstly, the user can select the translational DOFs to be considered for the candidate nodes in the OSP (see Fig. 2). By default,  $U_x$ ,  $U_y$ , and  $U_z$  are preselected for three-dimensional problems. If a two-dimensional problem is analyzed, only one or two of the three translational DOFs should be selected. Next, the target mode shapes can be specified by selecting the corresponding checkboxes. Finally, the desired OSP algorithm can be chosen from the drop-down menu, and the OSP problem is executed by clicking the *Run* button.



### 2.2.4. Visualize and export results

Once the OSP problem is executed, the GUI undergoes several changes:

- The OSP convergence curve (minimum value of  $E_d$  versus number of sensors) is displayed in the **Visualization Panel**. The term  $E_d$  represents the relative contribution of the DOFs to the rank of the Fisher Information Matrix in the EFI algorithm, or the corresponding weighted matrix used by the other algorithms (refer to [20] for further information).
- The *OSP DOFs* button appears, and the *Generate Report* and *Export to CAD* buttons are activated.

The sensor locations can be visualized in the **Visualization Panel** by selecting *Geometry* from the drop-down menu in the upper-left section. Clicking the *OSP DOFs* button enables additional options. The number of sensors, which is by default set to the number of target modes (the minimum required by the implemented OSP algorithms), can be adjusted using the drop-down menu. The selected value is reflected in the convergence curve as a vertical red line. Additional information about the selected DOFs is displayed in a table within the **OSP Panel** section. This table contains as many rows as the number of selected sensors and displays the corresponding DOFs, sorted in descending order of their associated  $E_d$  values. The columns are:

- **“Reference”**: Indicates whether the DOF was defined as a reference in the SAP2000 model (1 if true, 0 otherwise).
- **“ $E_d$ ”**: Displays the  $E_d$  value of the selected DOFs as computed by the OSP algorithm.
- **“OSP”**: Shows the node label in SAP2000 along with its corresponding measurement direction.

Additionally, if a G-OSP analysis has been performed, the *Generate References* button and the *References* drop-down menu become available. These options allow the creation of a new SAP2000 (.sdb) model prepared for an M-OSP analysis (i.e., with a load pattern named ‘References’), as previously explained in Section 2.2.1). The desired number of reference sensors can be set via the *N references* drop-down menu. The considered reference sensors are those with the highest  $E_d$  values, corresponding to the first elements listed in the previously mentioned table.

Finally, the *Export to CAD* and *Generate Report* buttons generate a .dxf and a .docx report, respectively. The generated .dxf file includes the SAP2000 model geometry with sensor positions marked by labeled arrows, organized in different layers according to the defined setups. On the other hand, the .docx report consists of two main sections:

- **SAP2000 Model:**
  - **Geometry**: Figures depicting the model’s geometry.
  - **Target Mode Shapes**: Images illustrating the selected target mode shapes.
- **OSP results:**
  - **Selected DOFs**: A table listing the selected DOFs for sensor placement (identical to the one in the *OSP Settings* section).
  - **OSP performance**: Visual representations of the Auto Modal Assurance Criterion (Auto-MAC) for the target modes, both for the selected DOFs and all DOFs in the model.
  - **Sensor locations**: Figures showing the optimal sensor locations obtained by the OSP algorithm.

## 3. Illustrative examples

This section demonstrates the application of OSP-SAP to two FEMs generated using the default templates in SAP2000 for easy reproducibility. Both case studies are included in the accompanying repository.

### 3.1. Case study I: G-OSP of a continuous beam

This example demonstrates the use of OSP-SAP for G-OSP of a three-span steel beam. The goal is to determine the optimal placement of four sensors only using the  $U_z$  DOF, which is suitable for identifying vertical bending modes around its strong axis.

The model is created in SAP2000 using the *Beam* template, consisting of three 6-meter spans. To facilitate result reproducibility, the default I-shaped cross-section and S355 steel material are used. After generating the initial model, each span is uniformly divided into 10 elements, resulting in 9 equally spaced internal nodes per span. Then, the node group ‘Candidate\_DOFs’ is defined, including all model nodes. Finally, the FEM is exported to a .xls database and further imported into OSP-SAP.

This model is three-dimensional, exhibiting bending modes in both vertical and transversal directions. However, since only vertical modes are of interest, only the  $U_z$  DOF is selected in OSP-SAP. Among the 12 mode shapes considered in the modal analysis, only the vertical modes (modes 4, 5, and 9) are selected as target modes. In this example, the SEMRO algorithm is executed. After running the algorithm, the optimal sensor locations appear in the GUI’s **Visualization Panel**, as shown in Fig. 6(a).

### 3.2. Case study II: M-OSP of a three-dimensional multi-storey building

This second case study focuses on a multi-storey building and demonstrates the use of OSP-SAP for defining reference sensors and conducting multi-setup OSP (M-OSP). The process is organized in two stages:

1. **Stage 1—G-OSP**: Solve the G-OSP problem to determine the positions of the reference sensors.
2. **Stage 2—M-OSP**: Determine the optimal locations of roving sensors for each setup considering the previously identified reference sensors.

The model is created in SAP2000 using the *3D Frames* template. It consists of three stories, each 3 m high, and five bays in the longitudinal direction and two in the transversal direction of the building, each 6 m long. As in the previous case study, the default beam I-shaped cross-section and S355 steel are utilized. Next, two groups named ‘Setup\_1’ and ‘Setup\_2’ are defined, required for conducting an M-OSP. The first setup contains the nodes of the first two bays in the longitudinal direction, while the second setup contains the remaining nodes.

In the OSP-SAP GUI, after importing the .xls database, modal analysis is conducted considering the first eight modes, and the EFI OSP algorithm is chosen. Unlike the previous case study, there is no need to define the ‘Candidate\_DOFs’ group in SAP2000, as the software automatically considers the nodes in ‘Setup\_1’ and ‘Setup\_2’ as candidates. Moreover, since the *References* load pattern is not defined, the software defaults to solving a G-OSP problem (as illustrated in Fig. 4). Once the algorithm finishes, users can navigate to the *OSP DOFs* tab, set the number of references (in this case two), and click on *Generate References* to obtain the .sdb model required for the M-OSP case.

The generated .sdb model must then be opened in SAP2000 and saved as .xls. With the updated model containing reference sensors, the M-OSP problem can be solved. After running the modal analysis,

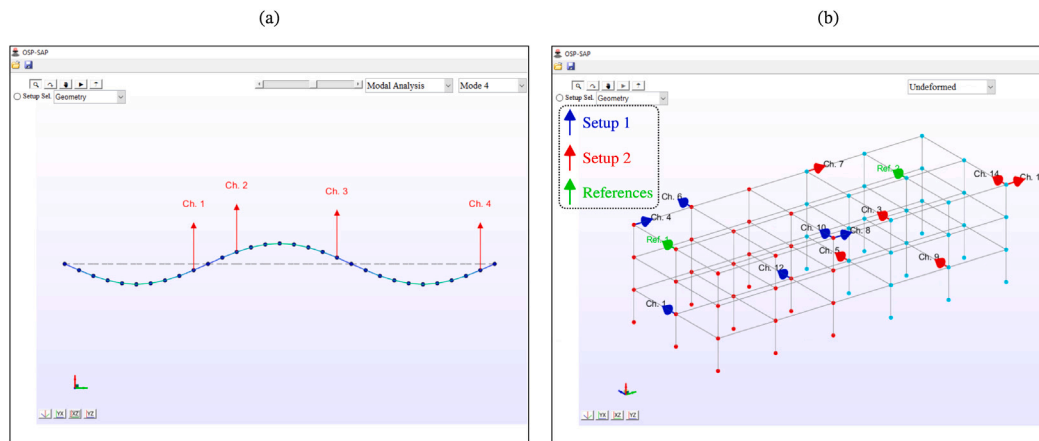


Fig. 6. Screenshot of OSP-SAP showing the OSP results for Case Study I (a) and Case Study II (b).

the EFI algorithm can be chosen. When the *Run* button is clicked, a prompt will appear allowing the user to select *Run Multisetup-OSP*. The resulting OSP is shown in Fig. 6(b), where arrows of different colors indicate the optimal positions of both the roving and reference sensors.

#### 4. Impact

This software provides an efficient and accessible tool for the scientific and engineering community to automatically implement and compare the performance of four widely recognized OSP algorithms: EFI, EFIwm, KEMRO, and SEMRO. Beyond facilitating OSP research, OSP-SAP presents practical advantages over existing solutions. Unlike previous tools requiring custom input formats, OSP-SAP directly imports SAP2000 models, eliminating the need for time-consuming model conversions and ensuring compatibility with this widely used structural analysis software. Its intuitive GUI enables an interactive and visual interpretation of optimal sensor configurations without requiring deep expertise in numerical optimization or programming. Furthermore, the open-source scripts, thoroughly commented, allow the scientific community to expand the software's capabilities. Practitioners managing modal identification tests utilizing OSP-SAP will benefit from OSP advantages, leading to cost reductions, elimination of redundant data, and improved data processing. This in turn will contribute to the deployment of SHM technologies in practical engineering scenarios.

#### 5. Conclusions

OSP-SAP features a user-friendly GUI that facilitates the implementation of widely accepted and tested optimal sensor placement algorithms. The software integrates with SAP2000, providing intuitive visualization and export features that enhance accessibility to researchers and practitioners. The software is available in an open repository with the aim of becoming a valuable tool for the SHM community. Future work may include the integration of additional OSP algorithms, considering advanced OSP criteria such as mutual information or entropy, to better account for both model and measurement uncertainties. The implementation of advanced features — such as tools to compare the outputs of different algorithms — could provide users with a deeper understanding of the software's underlying mechanics. Extending compatibility with cloud-based SHM systems, as well as with other FEM software such as ABAQUS and ANSYS, may also be considered.

#### CRediT authorship contribution statement

**Antonio S. López-Cuervo:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Enrique García-Macías:** Writing – review & editing, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rafael Castro-Triguero:** Investigation, Formal analysis, Conceptualization. **Juan Chiacchio:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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During the preparation of this work, the authors made use of ChatGPT to assist with reviewing the writing of the manuscript, as well as to improve the clarity and formatting of the supplementary documentation included in the GitHub repository. Following its use, the authors thoroughly reviewed and edited the content as needed, and assume full responsibility for the content of the publication.

#### Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.softx.2025.102216>.

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